```
close all;
run(fullfile(fileparts(fileparts(mfilename('fullpath'))),
'setup_project.m'));
load(fullfile(projectRoot, 'Part1', 'part1_results.mat'));
% additional G5 is created with a slightly larger radius than G4
% to compare the performance of the networks in terms of rendezvous time as
required.
G5 = make_disk(P0, rmin+0.4);
WAs = {make_WA(G1), make_WA(G2), make_WA(G3), make_WA(G4), make_WA(G5)};
WBs = {make_WB(G1), make_WB(G2), make_WB(G3), make_WB(G4), make_WB(G5)};
n = size(P0,1);
Gs = {G1, G2, G3, G4, G5};
Wchecks(WAs, WBs, Gs);
% Where WAs and WBs are the weight matrices for the five network topologies,
% and G1, G2, G3, G4, G5 are the corresponding adjacency matrices.
% The function Wchecks verifies that the weight matrices satisfy certain
properties,
% such as row sums equal to 1 and non-negativity.

%Experiment E1: Assess the performance of networks 1-5 for fastest rendevous.
% (Bradley and Alex please expand on this I don't fully understand the
topologies)
%The initial positions of the robots are given by P0, and the weight
matrices for each network topology are given by WAs and WBs.
% Predict broadcast protocal WA1 and WB1 will be the fastest because all the
robots will be able to communicate with each other directly,
% allowing for faster convergence to a common position.
% I predict that star will also be quite fast becaus the most robots will
quickly converge to the center of the star, as they can all communicate with
the center robot.
% The ring topology will be slower because the robots can only communicate
with their immediate neighbors, which will slow down the convergence process.
% The disk topology will also be slower because the robots can only
communicate with other robots within a certain distance,
% which may limit the number of robots that can communicate with each other
at any given time,
% especially if there are outlying robots that are far from others.

% Setting rendezvous condition to be 1% of the spread of the initial
positions of the robots,
% as instructed by the handout.
rendezvous_condition = 0.01*Spread(P0(:,1),P0(:,2));
rendevous_times = zeros(5,2);

for i = 1:5
    % Looping through each of the topologies
    K=7000;
    %Set the initial number of time steps to simulate to 7000.
    break_flag = false;
    % break_flag is used later to exit the while loop

    WA = WAs{i};
```

```
    WB = WBs{i};
    X0 = P0(:,1);
    Y0 = P0(:,2);
    X_A = simulate_linear(WA, X0, K);
    Y_A = simulate_linear(WA, Y0, K);
    X_B = simulate_linear(WB, X0, K);
    Y_B = simulate_linear(WB, Y0, K);


    for k = 1:K+1
        spread_A(k) = Spread(X_A(:,k),Y_A(:,k));
        %Calling the spread function (delta_k in the hand out) to compute
the spread of robots
        spread_B(k) = Spread(X_B(:,k),Y_B(:,k));
        if spread_A(k) < rendezvous_condition && spread_B(k) <
rendezvous_condition
            break_flag = true;
            %While loop will break if both networks meet the rendezvous
condition at the same time step k.
        end
        if spread_A(k) < rendezvous_condition && rendevous_times(i,1) == 0
            rendevous_times(i,1) = k;
            %update the rendezvous time for network A if it meets the
condition and has not been recorded yet.
        end
        if spread_B(k) < rendezvous_condition && rendevous_times(i,2) == 0
            rendevous_times(i,2) = k;
            %update the rendezvous time for network B if it meets the
condition and has not been recorded yet.
        end
    end

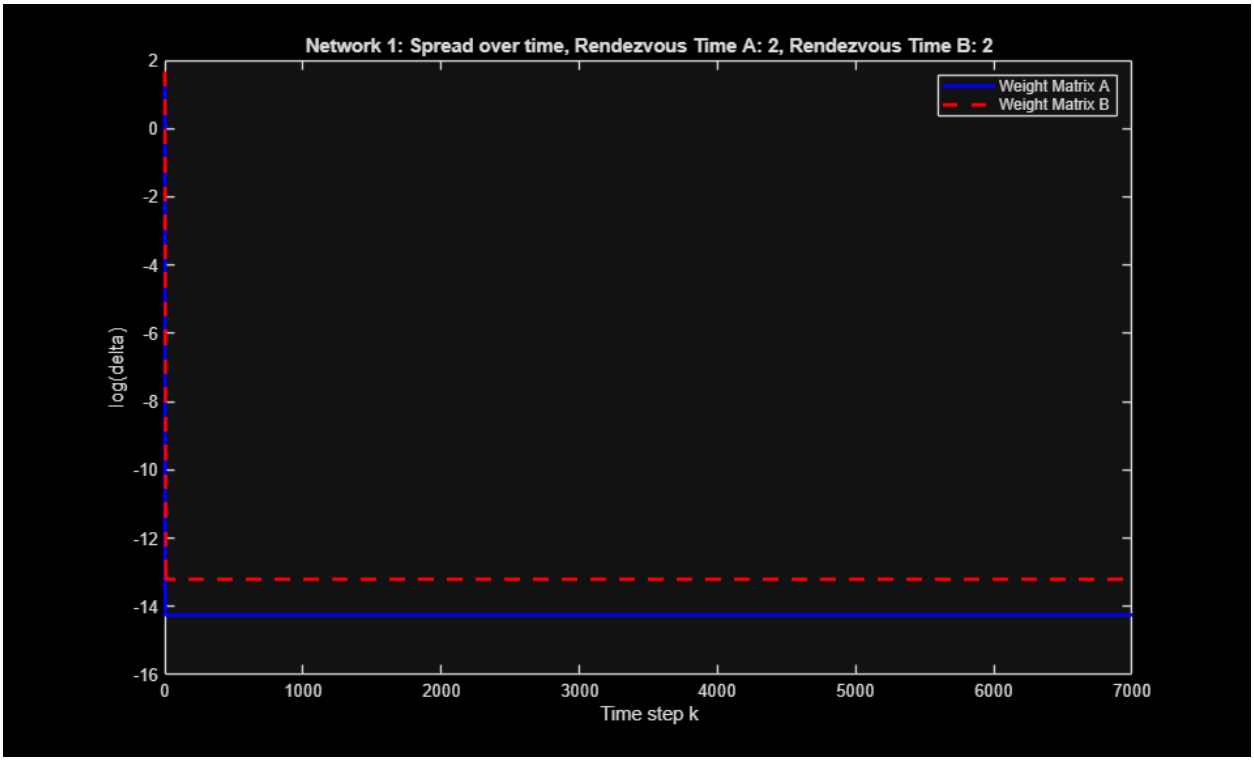
    if i == 2
        star_WA_final = [mean(X_A(:, 7000)), mean(Y_A(:, 7000))];
        star_WB_final = [mean(X_B(:, 7000)), mean(Y_B(:, 7000))];
    end

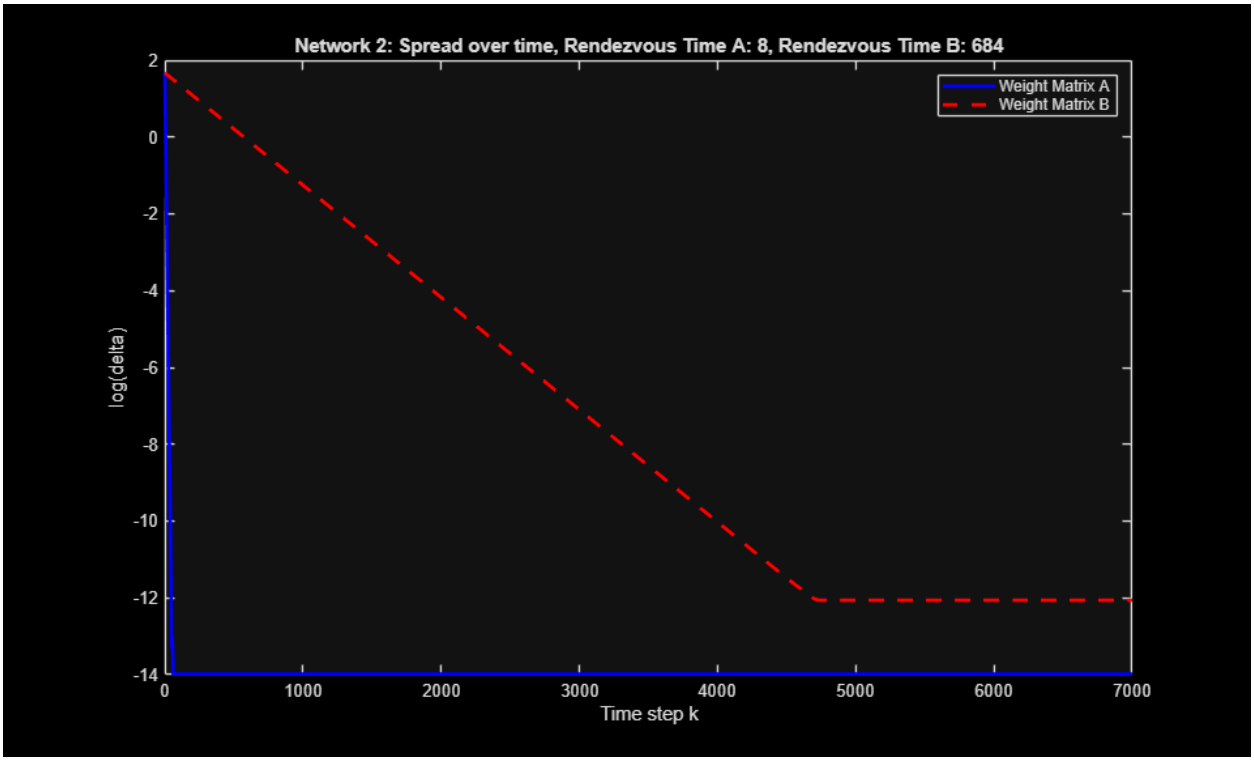
    figure;
    t = (0:length(spread_A)-1).';
    plot(t, log10(spread_A), 'b-', 'LineWidth', 2); hold on;
    plot(t, log10(spread_B), 'r--', 'LineWidth', 2); hold off;
    xlabel('Time step k'); ylabel('log(delta)');
    title(sprintf('Network %d: Spread over time, Rendezvous Time A: %d,
Rendezvous Time B: %d', i, rendevous_times(i,1), rendevous_times(i,2)));
    legend('Weight Matrix A', 'Weight Matrix B');
    %Plotting, nothing to see here :)

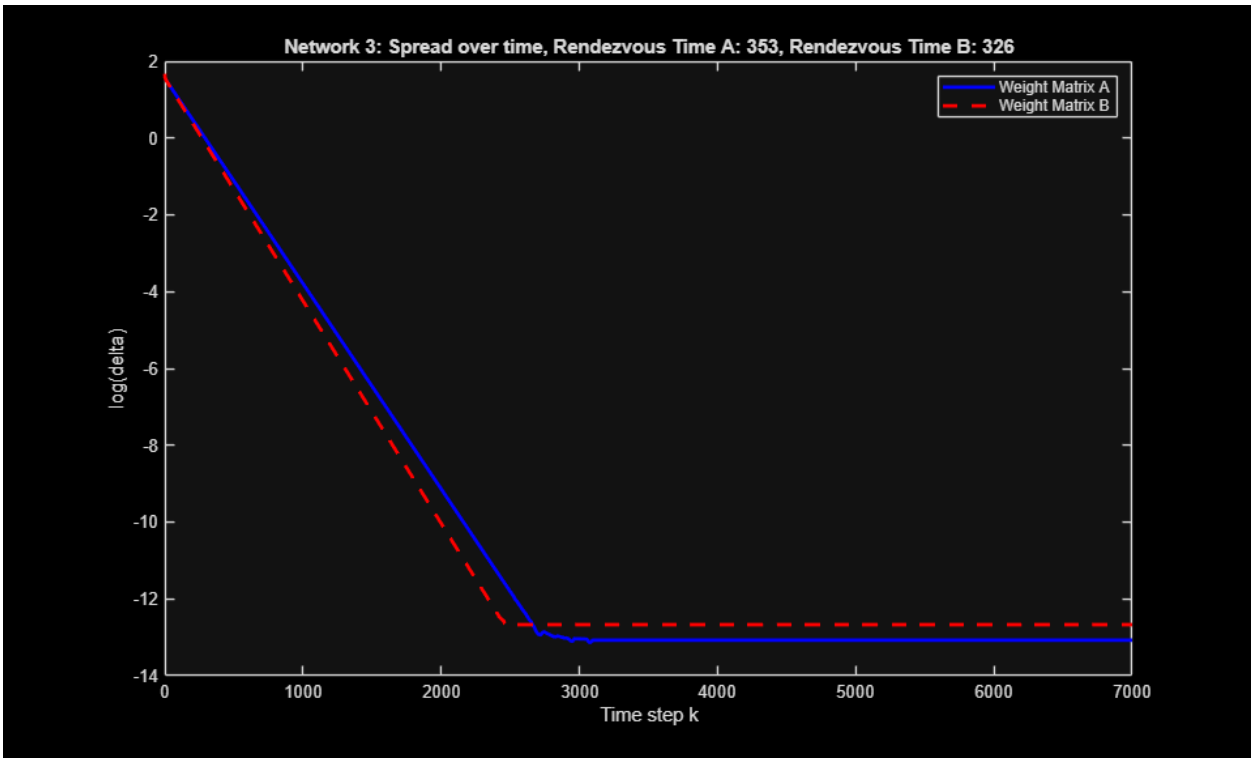
end
predicted = [mean(X0), mean(Y0)];
E2difference = [norm(star_WA_final - predicted), norm(star_WB_final -
predicted)];
% WB reaches rendezvous faster
```

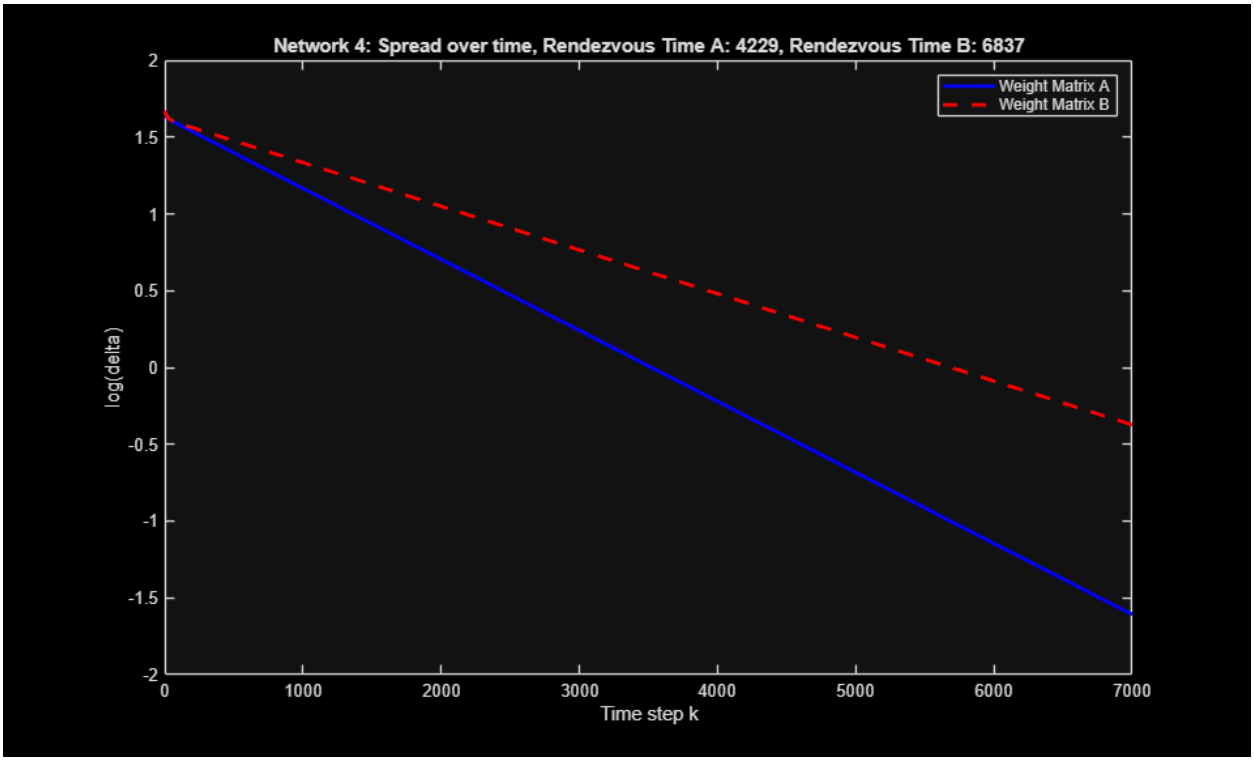
```
save(fullfile(projectRoot, 'Part2', 'part2_results.mat'), 'K');
```

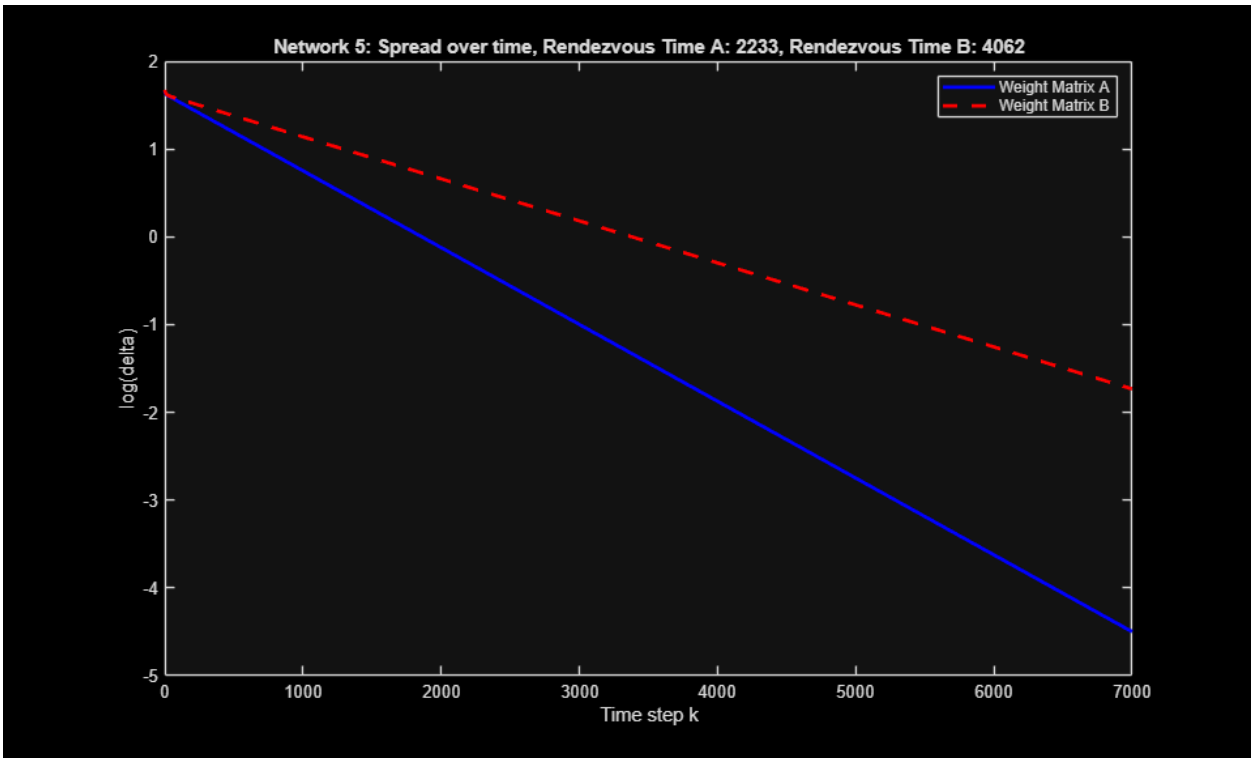
All W checks passed




Network 2: Spread over time, Rendezvous Time A: 8, Rendezvous Time B: 684
Weight Matrix A
Weight Matrix B
log(delta)
Time step k
2
0
-2
-4
-6
-8
-10
-12
-14
0
1000
2000
3000
4000
5000
6000
7000


Network 3: Spread over time, Rendezvous Time A: 353, Rendezvous Time B: 326
Weight Matrix A
Weight Matrix B
log(delta)
Time step k
2
0
-2
-4
-6
-8
-10
-12
-14
0
1000
2000
3000
4000
5000
6000
7000





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